Experimental Methods in Solid State Physics

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Winter Term 2025

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1 Reciprocal space probes: theory of diffraction

Historical Background

The discovery of X-ray diffraction is deeply rooted in Munich's scientific tradition. At the beginning of the 20th century, the city was one of Europe's centers for theoretical physics under the leadership of **Arnold Sommerfeld** at the *Ludwig-Maximilians-Universität* (LMU). Sommerfeld was known for his unusual philosophy that theoretical understanding and hands-on experimentation must go hand in hand and, therefore, a theory chair should also host experimental work . His institute therefore housed not only mathematicians and theorists but also instruments for optical and X-ray studies.

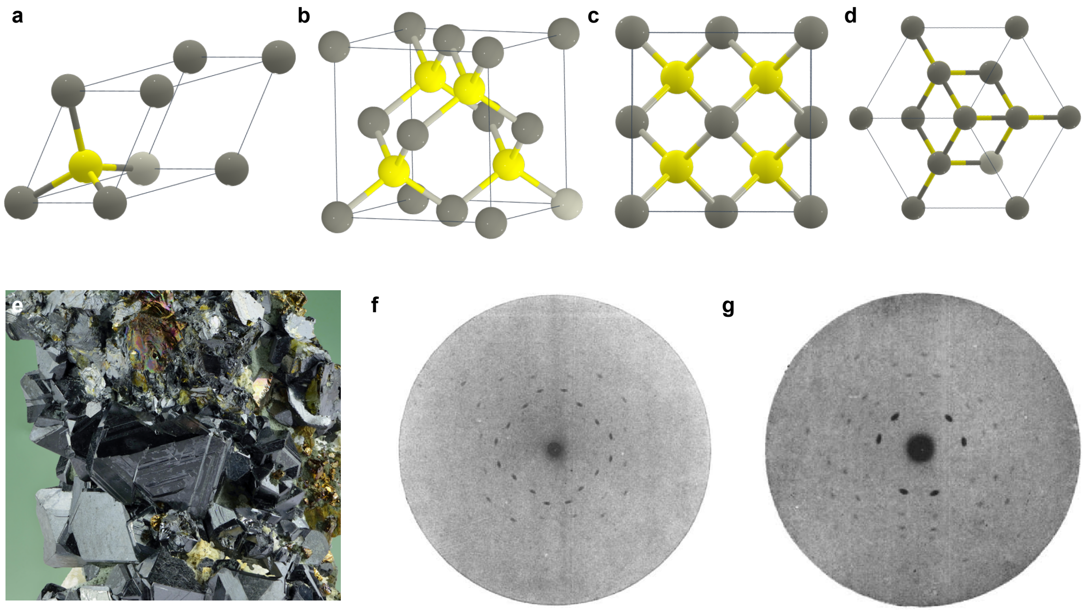

Figure 1: Laue's X-ray diffraction experiment on zinc blende (ZnS). **a** Primitive unit cell with one zinc and one sulfur atom in the basis. **b** Conventional fcc unit cell. **c** Projection along the (100) direction showing fourfold symmetry. **d** Projection along the (111) direction showing threefold symmetry. **e** Natural zinc blende crystals (sphalerite), showing facets of different orientations. Reproduced under CC BY-SA 4.0, I. Leidus 2021. **f,g** Original Laue X-ray diffraction patterns of zinc blende (ZnS) for incidence along the [001] (fourfold) and [111] (threefold) crystallographic directions. These directions are normal to the (001) and (111) plane families, respectively, which give rise to the characteristic square and triangular symmetries observed in the diffraction patterns. Reproduced with permission from W. Friedrich *et al.*, *Ann. Phys.* **41**, 971–988 (1913). Copyright Wiley.

In this environment, **Max von Laue**, then a young Privatdozent working with Sommerfeld, developed the idea that X-rays could be diffracted by a crystal. Stimulated by discussions with **Paul Peter Ewald**, who analyzed the propagation of waves in periodic media, Laue realized that a crystal lattice could act as a three-dimensional diffraction grating if the X-ray wavelength was comparable to interatomic spacings. In 1912, together with **Walter Friedrich** and **Paul Knipping**, Laue tested this hypothesis at LMU by directing X-rays onto a single crystal of zinc blende (ZnS). The resulting array of sharp spots — the first X-ray diffraction pattern — provided direct evidence that X-rays behave as waves and that atoms in crystals are arranged in a periodic lattice:

> "Diese Tatsache, daß eine völlige Vierzähligkeit auf der Platte vorhanden ist, ist wohl einer der schönsten Beweise für den gitterartigen Aufbau der Kristalle, und ein Beweis dafür, dass keine andere Eigenschaft als allein das Raumgitter hier in Betracht kommt."[1]

Laue's Diffraction Experiment on Zinc Blende

Zinc blende crystallizes in the face-centered cubic (fcc) structure with a two-atom basis, one zinc and one sulfur atom displaced along the body diagonal (Figs. 1**a,b**). Because of its high cubic symmetry, the crystal exhibits several distinct rotational axes, including a **fourfold axis** along the [001] direction and a **threefold axis** along the [111] direction (Figs. 1**c,d**). These correspond to the normals of the (100) and (111) planes, respectively.

In the Munich experiment, X-rays were aligned along these symmetry axes. For incidence along the fourfold [001] axis, the diffraction pattern displayed square symmetry (Fig. 1**f**); along the threefold [111] axis, the pattern revealed triangular symmetry (Fig. 1**g**). These symmetries reflected the orientation of reciprocal-lattice points in the respective crystal zones and provided direct evidence of the crystal's three-dimensional periodicity and the existence of a reciprocal lattice. For this discovery, Max von Laue received the *Nobel Prize in Physics in 1914*. Thus, the foundations of modern crystallography were laid in Munich through the synergy of theory and experiment fostered by Sommerfeld's school. Laue's experiment and the Braggs' interpretation remain cornerstones of solid-state physics.

Diffraction Geometry: From Bragg Reflection to Laue's Condition

Bragg Reflection from Lattice Planes. Very soon after the discovery of Laue diffraction, William Henry Bragg and his son William Lawrence Bragg proposed a simple and intuitive interpretation of how X-rays scatter from crystals. They viewed a crystal as a three-dimensional array of parallel lattice planes separated by a spacing d. When X-rays of wavelength λ are incident at an angle θ, waves reflected from adjacent planes interfere constructively if their path difference equals an integer multiple of λ (Fig. 2**a**).

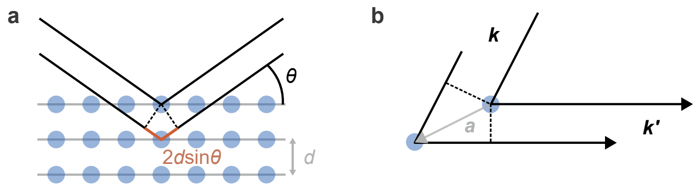

Figure 2: **Bragg vs. Laue. a** Geometry of Bragg reflection. Waves reflected under an angle θ from adjacent lattice planes with spacing d interfere constructively when the extra path $2d\sin\theta$ equals an integer multiple of λ. **b** Laue's condition. Each atom in the lattice acts as a scattering center. Constructive interference between an incoming wave $\mathbf{k}$ and outgoing wave $\mathbf{k}'$ occurs if for any lattice vector $\mathbf{R}$ the waves are in phase again, such that $(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R} = 2\pi n$.

[1] W. Friedrich, P. Knipping, M. von Laue, *Interferenzerscheinungen bei Röntgenstrahlen*, *Annalen der Physik*, **346**, 971–988 (1913).

From geometry, the path difference between rays reflected by neighboring planes is

$$\Delta s = 2d\sin\theta.$$

The condition for constructive interference is

$$2d\sin\theta = n\lambda, \qquad n = 1, 2, 3, \ldots$$

This is **Bragg's law**. It defines the angular positions of the diffraction maxima for a given lattice spacing d and wavelength λ. Each set of lattice planes (hkl) produces reflections at characteristic angles θ_{hkl}, providing the basis for indexing diffraction peaks in X-ray experiments.

Indexing of Directions and Planes

A crystal lattice is defined by the primitive translation vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$; any lattice point is at $\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$ with $n_i \in \mathbb{Z}$. These vectors are not generally orthogonal; the interaxial angles are $\alpha = \angle(\mathbf{a}_2, \mathbf{a}_3)$, $\beta = \angle(\mathbf{a}_1, \mathbf{a}_3)$, $\gamma = \angle(\mathbf{a}_1, \mathbf{a}_2)$.

Directions. A **crystallographic direction** is given by $\mathbf{d} = u\mathbf{a}_1 + v\mathbf{a}_2 + w\mathbf{a}_3$ and written as $[u\,v\,w]$. All symmetry-equivalent directions form the family $\langle u\,v\,w\rangle$.

Planes. A **crystal plane** intersects the axes at $(p\,\mathbf{a}_1, q\,\mathbf{a}_2, r\,\mathbf{a}_3)$. The reciprocals of these intercepts define the **Miller indices** (hkl) with $h = 1/p$, $k = 1/q$, $\ell = 1/r$. Parentheses (hkl) denote one plane, braces $\{hkl\}$ the equivalent family.

Relation. Directions $[uvw]$ belong to the *direct lattice*; plane normals (hkl) correspond to vectors of the *reciprocal lattice*. Hence $[hkl] \not\perp (hkl)$ in general. Only for cubic crystals, where $\mathbf{a}_i \perp \mathbf{a}_j$ and $|\mathbf{a}_i|$ are equal, are $[hkl]$ and (hkl) parallel.

Plane Spacing. For cubic symmetry, the interplanar spacing is $d_{hkl} = a/\sqrt{h^2 + k^2 + \ell^2}$, entering Bragg's law $2d_{hkl}\sin\theta = n\lambda$. Each diffraction maximum can thus be **indexed** by its (hkl) triplet.

Laue's Diffraction Condition. In a more general approach, one assumes that the incoming wave is scattered from all atoms in the crystal. The scattered, outgoing waves interfere with one another and produce a diffraction pattern. For a well-defined diffraction maximum, the waves scattered from *every* pair of atoms must be in phase. The path difference is the sum of the highlighted paths in Fig. 2**b**, whereby each can be expressed as the projection of $\mathbf{R}$ on the incoming and outgoing directions $\mathbf{k}/|k|$ and $\mathbf{k}'/|k'|$

$$\Delta s = \mathbf{R} \cdot \frac{-\mathbf{k}}{|k|} + \mathbf{R} \cdot \frac{\mathbf{k}'}{|k'|} = \mathbf{R} \cdot \left(\frac{\mathbf{k}'}{|k'|} - \frac{\mathbf{k}}{|k|}\right).$$

Making use of the fact that the scattering process is elastic $|k| = |k'|$ and $k = 2\pi/\lambda$, we arrive at the following condition for constructive interference

$$\frac{\mathbf{R}}{k} \cdot (\mathbf{k}' - \mathbf{k}) = \frac{\mathbf{R}\lambda}{2\pi} \cdot (\mathbf{k}' - \mathbf{k}) = n\lambda \Longrightarrow \mathbf{R} \cdot (\mathbf{k}' - \mathbf{k}) = 2\pi n.$$

In short, if the atoms are separated by a lattice translation $\mathbf{a}$, the phase difference between their scattered waves is

$$\Delta\phi = \Delta\mathbf{k} \cdot \mathbf{R}, \qquad \Delta\mathbf{k} = \mathbf{k}' - \mathbf{k}.$$

Constructive interference requires that the phase be an integer multiple of 2π:

$$\Delta\mathbf{k} \cdot \mathbf{R} = 2\pi n \Longleftrightarrow e^{i\,\Delta\mathbf{k}\cdot\mathbf{R}} = 1.$$

This condition must hold for **any lattice translation** $\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$, not just for a few selected planes. A diffraction peak is therefore only well defined if the interference is constructive for all atoms separated by an arbitrary lattice vector. Because the three primitive vectors $\mathbf{a}_i$ form a linearly independent basis, this is the case if, and only if, the condition is satisfied for each of them individually:

$$e^{i\,\Delta\mathbf{k}\cdot\mathbf{a}_i} = 1, \qquad i = 1, 2, 3.$$

or equivalently

$$\Delta\mathbf{k} \cdot \mathbf{a}_i = 2\pi n_i, \qquad n_i \in \mathbb{Z}.$$

These three independent equations define the set of all $\Delta\mathbf{k}$ vectors that produce constructive interference — the discrete **lattice of allowed diffraction vectors**.

Bragg's law is the special case where (h, k, ℓ) refer to a single family of lattice planes with spacing d_{hkl}, yielding the same angular condition $2d_{hkl}\sin\theta = n\lambda$. In general, the integer triplet (h, k, ℓ) provides the natural way to **index diffraction peaks** in any X-ray diffraction pattern.

The Reciprocal Lattice and Ewald's sphere

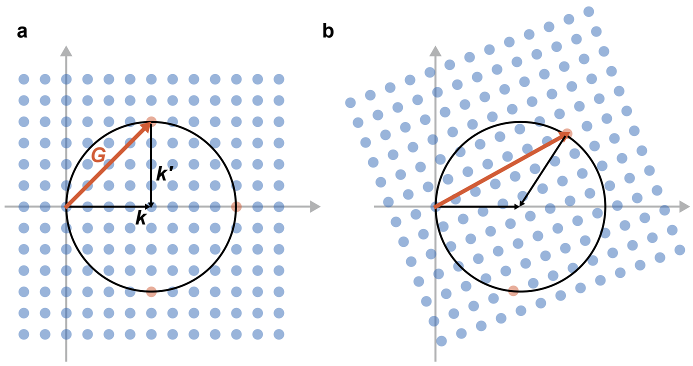

Figure 3: Ewald-sphere construction for two crystal orientations. **a** The incident wavevector $\mathbf{k}$ points from the origin to the center of the sphere (radius $2\pi/\lambda$). The blue dots are reciprocal lattice points. The grey axes denote directions in the laboratory frame where the diffracted beams are observed. A reciprocal-lattice point lying on the sphere yields a diffracted wavevector $\mathbf{k}' = \mathbf{k} + \mathbf{G}$. **b** After rotating the crystal, the reciprocal lattice rotates and a different point intersects the sphere, producing a new $\mathbf{k}'$.

The vectors $\Delta\mathbf{k}$ satisfying the Laue condition $e^{i\,\Delta\mathbf{k}\cdot\mathbf{a}} = 1$ form a periodic lattice in wave-vector space, the so-called **reciprocal lattice**.aMthematically, the reciprocal basis vectors $\mathbf{b}_i$ are defined so that $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$. Since the $\mathbf{a}_i$ are linearly independent, this relation uniquely determines the $\mathbf{b}_i$. This is the definition of a **dual basis** spanning a **dual space**. Each allowed diffraction vector $\mathbf{G} = h\mathbf{b}_1 + k\mathbf{b}_2 + \ell\mathbf{b}_3$ then automatically fulfills the Laue condition. The reciprocal lattice therefore provides a compact and elegant framework to describe the periodicity of a crystal in $\mathbf{k}$-space and to formulate the diffraction condition in terms of reciprocal lattice vectors $\mathbf{G} = \mathbf{k}' - \mathbf{k}$. The

Ewald sphere provides a geometrical construction of possible scattering vectors for specific incident directions, wavelength, and crystal orientations (Fig. 3).

Reciprocal Lattice

The reciprocal lattice provides a compact description of the periodicity of a crystal in wavevector space. It follows directly from the Laue interference condition $e^{i\,\Delta\mathbf{k}\cdot\mathbf{a}_i} = 1$, which implies $\Delta\mathbf{k}\cdot\mathbf{a}_i = 2\pi n_i$ for all primitive vectors $\mathbf{a}_i$.

Definition. The reciprocal basis vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ are defined by

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\,\delta_{ij}.$$

Any reciprocal-lattice vector is

$$\mathbf{G}_{hkl} = h\,\mathbf{b}_1 + k\,\mathbf{b}_2 + \ell\,\mathbf{b}_3, \qquad h, k, \ell \in \mathbb{Z}.$$

Each $\mathbf{G}_{hkl}$ corresponds to a periodic modulation that repeats in phase after every lattice translation.

Explicit construction. For a direct lattice spanned by $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ with cell volume $V = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)$,

$$\mathbf{b}_1 = 2\pi\,\frac{\mathbf{a}_2 \times \mathbf{a}_3}{V}, \quad \mathbf{b}_2 = 2\pi\,\frac{\mathbf{a}_3 \times \mathbf{a}_1}{V}, \quad \mathbf{b}_3 = 2\pi\,\frac{\mathbf{a}_1 \times \mathbf{a}_2}{V}.$$

The reciprocal cell volume is $V^* = (2\pi)^3/V$.

Geometric meaning. Each $\mathbf{G}_{hkl}$ is perpendicular to the family of planes (hkl), satisfying $\mathbf{G}_{hkl} \cdot \mathbf{r} = 2\pi n$. The spacing between adjacent planes is $d_{hkl} = 2\pi/|\mathbf{G}_{hkl}|$. In cubic crystals, $\mathbf{b}_i$ are parallel to $\mathbf{a}_i$ with $|\mathbf{b}_i| = 2\pi/a$.

Diffraction condition. Constructive interference (Laue condition) occurs when

$$\mathbf{k}' - \mathbf{k} = \mathbf{G}_{hkl}.$$

Thus, each diffraction peak corresponds to one reciprocal-lattice vector lying on the Ewald sphere of radius $|\mathbf{k}| = 2\pi/\lambda$.

Question 1 *Bragg Reflection*

An X-ray beam of wavelength $\lambda = 0.154$ nm is incident on a cubic crystal with lattice constant $a = 0.408$ nm. At which angle 2θ does the first-order Bragg reflection from the (111) planes occur?

(a) $14.5°$
(b) $25.9°$
(c) $37.1°$
(d) $44.5°$

Question 2 *Constructive Interference*

Which condition must be fulfilled for a crystal to produce sharp and well-defined diffraction peaks?

(a) The X-ray beam must be incident along a high-symmetry lattice direction.

(b) Constructive interference must occur for a few selected lattice planes only.

(c) Constructive interference must hold for all atoms separated by any lattice translation vector.

(d) Each atom must scatter with the same amplitude and phase.

Question 3 *Bragg–Laue Relation*

Which statement correctly describes the relationship between the Bragg and Laue formulations of diffraction?

(a) Bragg's law neglects the vector nature of the wavevectors and is only approximate.

(b) Both describe the same elastic interference condition; Bragg's law is a geometric projection of the Laue equations.

(c) Laue's formulation applies only for inelastic scattering.

(d) Bragg's law applies only to cubic crystals, while Laue's holds for all lattices.

Question 4 *Ewald Sphere Comparison*

For an X-ray beam and an electron beam of the *same kinetic energy*, which statement correctly explains the difference in their diffraction patterns?

(a) Both show the same number of diffraction spots because the Ewald sphere radius depends only on energy.

(b) Electrons have a much shorter wavelength at the same energy, giving a larger Ewald-sphere radius and many more reciprocal-lattice intersections.

(c) X-rays interact more strongly with the lattice, producing additional higher-order reflections.

(d) Electron diffraction is mostly inelastic, so many spots appear from different scattering processes.

2 X-ray Scattering and Crystal Structure Determination

Crystals are periodic arrangements of atoms and, therefore, of electron density. Any probe with wavelength comparable to the lattice spacing, $\lambda \sim 1\,\text{Å}$, will be diffracted by this periodic electron density. In principle, very different probes can be used: electromagnetic waves (X-rays) and matter waves (electrons and neutrons). All of them produce diffraction when their de Broglie wavelength matches the crystal periodicity, but they differ strongly in their interaction with matter and in the experimental techniques they require.

probe	interaction	surface/bulk	source	$E(\lambda = 1\,\text{Å})$
X-rays	electron density	bulk (mm)	table-top, synchrotron	$\approx 12.4\,\text{keV}$
electrons	Coulomb interaction	surface (nm)	electron gun	$\approx 150\,\text{eV}$
neutrons	nuclear & magnetic	bulk (cm)	reactor, spallation source	$\approx 0.082\,\text{eV}$

Table 1: Comparison of common diffraction probes with wavelength $\lambda = 1\,\text{Å}$.

In this chapter we focus on X-rays. X-ray diffraction (XRD) is the standard technique for determining lattice constants, crystal symmetry, and the arrangement of atoms within the unit cell. The key theoretical insight is that, in the limit of weak scattering, the diffracted amplitude is given by the Fourier transform of the electron density. The discrete diffraction peaks observed from a crystal are then simply the Fourier components of a periodic charge density. In the following sections we develop this Fourier-space description and connect it to the concepts of *atomic form factor* and *structure factor*, which control the intensities of the observed reflections.

Scattering as a Fourier Transform of the Electron Density

We consider an incident monochromatic X-ray beam with wavevector $\mathbf{k}$, described by a plane wave

$$\psi_{\text{in}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}.$$

After interaction with the sample, part of the wave is scattered into a new direction with wavevector $\mathbf{k}'$. Far from the sample, the scattered wave has the standard outgoing spherical form

$$\psi_{\text{out}}(\mathbf{r}) \propto A(\mathbf{k} \to \mathbf{k}')\,\frac{e^{ikr}}{r},$$

where $A(\mathbf{k} \to \mathbf{k}')$ is the scattering amplitude. It depends only on the difference between incoming and outgoing wavevectors,

$$\mathbf{q} = \mathbf{k}' - \mathbf{k}.$$

For X-rays, the interaction with matter is weak and primarily with the electron density $\rho(\mathbf{r})$. Within the first Born approximation (single scattering), one finds that the scattering amplitude is given by the Fourier transform of $\rho(\mathbf{r})$,

$$A(\mathbf{q}) = \int \rho(\mathbf{r})\, e^{-i\mathbf{q}\cdot\mathbf{r}}\, \mathrm{d}^3 r, \qquad \mathbf{q} = \mathbf{k}' - \mathbf{k}.$$

All overall prefactors (classical electron radius, polarization factors, detector efficiency, ...) are absorbed into the proportionality constant that relates the measured intensity to the scattering amplitude via

$$I(\mathbf{q}) \;\propto\; |A(\mathbf{q})|^2.$$

Born Approximation and the Green-Function Method

1. From Schrödinger to the inhomogeneous Helmholtz equation. For a particle of mass m in a scattering potential $V(\mathbf{r})$, the stationary Schrödinger equation can be rewritten as an inhomogeneous Helmholtz equation

$$(\nabla^2 + k^2)\psi(\mathbf{r}) = \frac{2m}{\hbar^2} V(\mathbf{r})\,\psi(\mathbf{r}), \qquad k^2 = \frac{2mE}{\hbar^2},$$

2. Outgoing Green function. The Green function of $(\nabla^2 + k^2)$ is defined by

$$(\nabla^2 + k^2)\, G^{(+)}(\mathbf{r}, \mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}').$$

The physical (outgoing) solution is a spherical wave

$$G^{(+)}(\mathbf{r}, \mathbf{r}') = -\frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r} - \mathbf{r}'|},$$

3. Lippmann–Schwinger equation. Multiply the Helmholtz equation for ψ by $G^{(+)}$, the Green equation by ψ, subtract, and integrate over all space. Using Gauss's theorem and outgoing boundary conditions yields the integral form of the scattering state:

$$\boxed{\psi(\mathbf{r}) = \psi_{\text{in}}(\mathbf{r}) + \frac{2m}{\hbar^2} \int G^{(+)}(\mathbf{r}, \mathbf{r}')\, V(\mathbf{r}')\, \psi(\mathbf{r}')\, d^3r'.}$$

This expresses the scattered wave as a superposition of outgoing spherical waves emitted from each point in the potential.

4. First Born approximation (weak scattering). If the potential is weak (X-rays interact only slightly with electrons), then inside the integral we approximate the full wave as the incident wave $\psi(\mathbf{r}') \approx e^{i\mathbf{k}\cdot\mathbf{r}'}$ (equivalent to first-order-perturbation theory) yielding

$$\psi(\mathbf{r}) \approx e^{i\mathbf{k}\cdot\mathbf{r}} + \frac{2m}{\hbar^2} \int G^{(+)}(\mathbf{r}, \mathbf{r}')\, V(\mathbf{r}')\, e^{i\mathbf{k}\cdot\mathbf{r}'}\, d^3r'.$$

5. Far-field limit and scattering amplitude. For $|\mathbf{r}| \to \infty$,

$$G^{(+)}(\mathbf{r}, \mathbf{r}') \approx -\frac{e^{ikr}}{4\pi r}\, e^{-i\mathbf{k}'\cdot\mathbf{r}'}, \qquad \mathbf{k}' = k\,\hat{\mathbf{r}}.$$

Thus the scattered wave has the standard asymptotic form

$$\psi(\mathbf{r}) \xrightarrow{r\to\infty} e^{i\mathbf{k}\cdot\mathbf{r}} + f(\mathbf{k}\to\mathbf{k}')\, \frac{e^{ikr}}{r},$$

with the Born amplitude

$$\boxed{f(\mathbf{k}\to\mathbf{k}') = -\frac{2m}{4\pi\hbar^2} \int V(\mathbf{r})\, e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\, d^3r.}$$

6. Application to X-ray scattering. For X-rays, interaction is governed by $V(\mathbf{r}) \propto \epsilon(\mathbf{r}) \propto \rho(\mathbf{r})$. Therefore, we define the scattering amplitude as

$$A(\mathbf{q}) = \int \rho(\mathbf{r})\, e^{-i\mathbf{q}\cdot\mathbf{r}}\, d^3r, \qquad \mathbf{q} = \mathbf{k}' - \mathbf{k},$$

where the integral runs across the sample volume, i.e. the scattered amplitude is the Fourier transform of the electron density.

From Electron Density to the Atomic Form Factor

The electron density of a solid can be written as a sum of atomic contributions located at the atomic positions $\mathbf{r}_\alpha$,

$$\rho(\mathbf{r}) = \sum_\alpha \rho_\alpha(\mathbf{r} - \mathbf{r}_\alpha),$$

where $\rho_\alpha(\mathbf{r})$ denotes the electron density associated with atom α. Note that we do not need to assume a periodic lattice at this point of the calculation. Using the definition of the scattering amplitude

$$A(\mathbf{q}) = \int \rho(\mathbf{r})\, e^{-i\mathbf{q}\cdot\mathbf{r}}\, \mathrm{d}^3 r,$$

we obtain

$$\begin{aligned} A(\mathbf{q}) &= \sum_\alpha \int \rho_\alpha(\mathbf{r} - \mathbf{r}_\alpha)\, e^{-i\mathbf{q}\cdot\mathbf{r}}\, \mathrm{d}^3 r \\ &= \sum_\alpha e^{-i\mathbf{q}\cdot\mathbf{r}_\alpha} \int \rho_\alpha(\mathbf{r})\, e^{-i\mathbf{q}\cdot\mathbf{r}}\, \mathrm{d}^3 r. \end{aligned}$$

The integral over the internal coordinate $\mathbf{r}$ depends only on the atomic electron density and defines the *atomic form factor*

$$f_\alpha(\mathbf{q}) = \int \rho_\alpha(\mathbf{r})\, e^{-i\mathbf{q}\cdot\mathbf{r}}\, \mathrm{d}^3 r.$$

Physical meaning. The atomic form factor $f_\alpha(\mathbf{q})$ describes how strongly a single atom scatters X-rays at momentum transfer $\mathbf{q}$. It is the Fourier transform of the atomic electron density and therefore decreases with increasing $|\mathbf{q}|$: high-angle scattering probes finer spatial features of the electron cloud, to which heavy atoms (large Z) contribute more strongly than light atoms. The phase factor $e^{-i\mathbf{q}\cdot\mathbf{r}_\alpha}$ accounts for the relative phase between waves scattered from different atoms; in an ordered crystal, these phases interfere and give rise to the *structure factor* introduced next.

Structure Factor and Interference from a Periodic Lattice

For a **periodic** crystal, the atomic electron density is repeated periodically at lattice vectors $\mathbf{R}$. For basis of atoms located at fixed positions $\mathbf{r}_\alpha$ within the unit cell, the position of atom j in the unit cell at lattice vector $\mathbf{R}$ is

$$\mathbf{R} + \mathbf{r}_\alpha.$$

The total electron density is therefore

$$\rho(\mathbf{r}) = \sum_{\mathbf{R}} \sum_\alpha \rho_\alpha(\mathbf{r} - \mathbf{R} - \mathbf{r}_\alpha),$$

and the corresponding scattering amplitude is a double sum over lattice points and basis atoms.

$$A(\mathbf{q}) = \sum_{\mathbf{R}} \sum_\alpha f_\alpha(\mathbf{q})\, e^{-i\mathbf{q}\cdot(\mathbf{R}+\mathbf{r}_\alpha)}.$$

This factorizes into

$$A(\mathbf{q}) = \left(\sum_{\mathbf{R}} e^{-i\mathbf{q}\cdot\mathbf{R}}\right) \left(\sum_\alpha f_\alpha(\mathbf{q})\, e^{-i\mathbf{q}\cdot\mathbf{r}_\alpha}\right).$$

The sum over lattice points,

$$S(\mathbf{q}) = \sum_{\mathbf{R}} e^{-i\mathbf{q}\cdot\mathbf{R}},$$

is nonzero *only* when $\mathbf{q}$ equals a reciprocal lattice vector $\mathbf{G}$. This is the basic diffraction condition, which we derived already in the last chapter based on Laue's or Bragg's arguments. Diffraction peaks occur only at reciprocal-lattice points:

$$\boxed{\mathbf{q} = \mathbf{G}}$$

Structure factor. For $\mathbf{q} = \mathbf{G}$, the remaining sum is called the *structure factor*.

$$F(\mathbf{G}) = \sum_{\alpha} f_\alpha(\mathbf{G})\, e^{-i\mathbf{G}\cdot\mathbf{r}_\alpha}.$$

Physical interpretation. Each atom contributes with its form factor $f_\alpha(\mathbf{G})$ weighted by a phase factor $e^{-i\mathbf{G}\cdot\mathbf{r}_\alpha}$ that depends on its position in the unit cell. Depending on the symmetry of the crystal, these contributions may interfere constructively or destructively. This explains why certain reflections are strong, weak, or completely absent in a diffraction pattern. The structure factor is therefore the central quantity connecting diffraction intensities to the crystal structure.

Example: Diffraction Spectrum of NaCl

Sodium chloride (NaCl) crystallizes in the rock-salt structure. Its Bravais lattice is face-centered cubic (fcc), which can be derived from a simple cubic lattice with 4 atoms in the basis (Fig. 4**a**). In the fcc lattive of NaCl there are two atoms (Cl, Na) per unit cell (Fig. 4**b**). Using this example, we will illustrate how lattice symmetry and basis interference together determine which diffraction peaks appear and with what intensity (Fig. 4**c**).

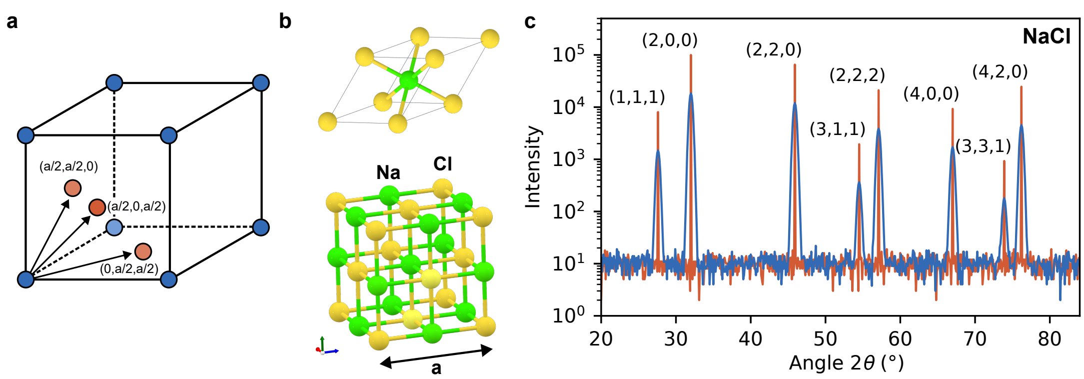

Figure 4: **Diffraction patterns of NaCl.** **a** Fcc lattice from simple cubic lattice with 4 atoms in the basis. **b** Primitive unit cell and conventional unit cell of NaCl. The lattice is fcc with two atoms in the basis: Cl located at $(0, 0, 0)$ and Na located at $a(1/2, 1/2, 1/2)$. **c** Simulated X-ray diffraction spectrum of NaCl assuming an average grain size of 1 nm (blue) and 100 nm (red). The fcc lattice allows only diffraction peaks with (hkl) all odd or even. The Na and Cl basis weakens the odd peaks due to partial destructive interference.

Conventional Cubic Cells and Miller Indexing in the Cubic Crystal Family

In crystallography the cubic crystal family (simple cubic, fcc, bcc) is always described using the *conventional cubic cell* of side length a, even though fcc and bcc lattices are not primitive in this basis. This convention is rarely stated explicitly, but it underlies all standard indexing using Miller indices (hkl).

Primitive vs. conventional cells. Every Bravais lattice has a primitive unit cell defined by three primitive vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$. For fcc and bcc these primitive cells are non-cubic and have volumes $a^3/4$ (fcc) and $a^3/2$ (bcc). Crystallography instead uses the larger *conventional cubic cell* with orthogonal edges of equal length a, together with a basis:

simple cubic:	1 atom basis,
bcc:	basis at $(0,0,0)$, $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$,
fcc:	basis at $(0,0,0)$, $(\frac{1}{2},\frac{1}{2},0)$, $(\frac{1}{2},0,\frac{1}{2})$, $(0,\frac{1}{2},\frac{1}{2})$.

Why the cubic cell is used. The conventional cubic cell preserves the *cubic metric*: the coordinate axes remain orthogonal and all edges have the same length. As a result, geometric properties of lattice planes are simple:

$$d_{hkl} = \frac{a}{\sqrt{h^2+k^2+\ell^2}}, \qquad \mathbf{G} = \frac{2\pi}{a}(h,k,\ell).$$

This makes Bragg's law and diffraction indexing intuitive and directly tied to experiments.

Selection rules as a basis effect. Because fcc and bcc Bravais lattices are represented in the cubic basis with a multi-point basis, not every cubic vector (hkl) corresponds to a reciprocal-lattice vector. The consistency between the cubic basis and the true primitive reciprocal lattice leads to the following extinction rules:

$$\text{fcc: } (hkl) \text{ all even or all odd}, \qquad \text{bcc: } h+k+\ell \text{ even}.$$

These rules arise entirely from the choice of the conventional cubic basis, not from the underlying Bravais lattice itself.

Summary. *The entire cubic crystal family is indexed in the conventional cubic basis rather than in the primitive basis.* This preserves the simplicity of cubic geometry for plane spacings and diffraction, while the multi-atom basis introduces the necessary selection rules for fcc and bcc crystals. This convention is universal in crystallography and X-ray diffraction.

Lattice Sum and Selection Rule for the fcc Lattice

We represent the fcc lattice as a simple cubic Bravais lattice with lattice constant a and a four-point basis. The simple cubic lattice vectors are

$$\mathbf{R} = a(n_1, n_2, n_3), \qquad n_1, n_2, n_3 \in \mathbb{Z},$$

and the four basis positions (in units of a) are

$$\mathbf{d}_1 = a(0,0,0), \quad \mathbf{d}_2 = a\Big(0, \tfrac{1}{2}, \tfrac{1}{2}\Big), \quad \mathbf{d}_3 = a\Big(\tfrac{1}{2}, 0, \tfrac{1}{2}\Big), \quad \mathbf{d}_4 = a\Big(\tfrac{1}{2}, \tfrac{1}{2}, 0\Big).$$

The set of all positions $\mathbf{R} + \mathbf{d}_\alpha$ $(\alpha = 1, \ldots, 4)$ is exactly the fcc lattice.

Lattice sum. The total phase factor for scattering with momentum transfer $\mathbf{q}$ from all lattice points can be written as

$$S(\mathbf{q}) = \sum_{\mathbf{R}} \sum_{\alpha=1}^{4} e^{-i\mathbf{q}\cdot(\mathbf{R}+\mathbf{d}_\alpha)}.$$

This factorizes into a simple-cubic lattice sum and a basis sum:

$$S(\mathbf{q}) = \Big(\sum_{\mathbf{R}} e^{-i\mathbf{q}\cdot\mathbf{R}}\Big)\Big(\sum_{\alpha=1}^{4} e^{-i\mathbf{q}\cdot\mathbf{d}_\alpha}\Big).$$

The first sum is nonzero when $\mathbf{q}$ coincides with a reciprocal lattice vector of the simple cubic lattice,

$$\mathbf{G} = \frac{2\pi}{a}(h, k, \ell), \qquad h, k, \ell \in \mathbb{Z}.$$

We therefore set $\mathbf{q} = \mathbf{G}$ and focus on the remaining factor

$$B(hkl) = \sum_{\alpha=1}^{4} e^{-i\mathbf{G}\cdot\mathbf{d}_\alpha},$$

which determines whether a given (hkl) reflection is allowed or forbidden.

Basis sum. Using the basis positions defined above and $\mathbf{G} = \frac{2\pi}{a}(h, k, \ell)$, we find

$$\begin{aligned}
\mathbf{G}\cdot\mathbf{d}_1 &= 0, \\
\mathbf{G}\cdot\mathbf{d}_2 &= \frac{2\pi}{a}\Big(h\cdot 0 + k\cdot\tfrac{a}{2} + \ell\cdot\tfrac{a}{2}\Big) = \pi(k+\ell), \\
\mathbf{G}\cdot\mathbf{d}_3 &= \pi(h+\ell), \\
\mathbf{G}\cdot\mathbf{d}_4 &= \pi(h+k).
\end{aligned}$$

Hence

$$B(hkl) = 1 + e^{-i\pi(k+\ell)} + e^{-i\pi(h+\ell)} + e^{-i\pi(h+k)}.$$

Since $e^{-i\pi n} = (-1)^n$, this becomes

$$B(hkl) = 1 + (-1)^{k+\ell} + (-1)^{h+\ell} + (-1)^{h+k}.$$

Evaluation of the selection rule. Now consider different parities of (h, k, ℓ):

- **All even or all odd.** If h, k, ℓ are all even or all odd, then $h+k$, $h+\ell$, and $k+\ell$ are all even. Therefore $(-1)^{h+k} = (-1)^{h+\ell} = (-1)^{k+\ell} = 1$, and
$$B(hkl) = 1 + 1 + 1 + 1 = 4 \neq 0.$$
- **Mixed parity (some even, some odd).** If one index is even and two are odd (or vice versa), then among the sums $h+k$, $h+\ell$, $k+\ell$ exactly two are odd and one is even, such that
$$B(hkl) = 1 + (+1) + (-1) + (-1) = 0.$$

The full lattice sum $S(\mathbf{q})$, is nonzero only if the Miller indices are either all even or all odd:

$$\boxed{S(\mathbf{q}) \neq 0 \quad \Leftrightarrow \quad h, k, \ell \text{ all even or all odd.}}$$

This is the characteristic selection rule for the fcc lattice: reflections with mixed parity of (h, k, ℓ) (such as 100, 210, 221, ...) are systematically absent.

Structure factor of NaCl

Within the fcc unit cell, the two atoms occupy the positions

$$\mathbf{r}_{\mathrm{Cl}} = (0, 0, 0), \qquad \mathbf{r}_{\mathrm{Na}} = \frac{a}{2}(1, 1, 1).$$

The position of Na differs from Cl by a body-diagonal shift of half a simple-cubic lattice vector. Note that several choices of the NaCl basis are possible, e.g. $(0, 0, 0)$ and $(\frac{1}{2}, 0, 0)$, since all lead to the same crystal after applying the fcc Bravais translations. The canonical choice $(0, 0, 0)$ and $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is preferred because it preserves the full cubic symmetry, makes the fcc Bravais lattice evident, and places the second atom at the center of the cube, highlighting the octahedral coordination characteristic of the rock-salt structure. For reflections that satisfy the fcc condition, the total structure factor is

$$F_{hkl} = f_{\mathrm{Cl}}(\mathbf{G})\, e^{-i\mathbf{G}\cdot\mathbf{r}_{\mathrm{Cl}}} + f_{\mathrm{Na}}(\mathbf{G})\, e^{-i\mathbf{G}\cdot\mathbf{r}_{\mathrm{Na}}},$$

where f_{Cl} and f_{Na} are the atomic form factors. Since $\mathbf{r}_{\mathrm{Cl}} = (0, 0, 0)$, the first phase factor is unity. The second phase factor becomes

$$e^{-i\mathbf{G}\cdot\mathbf{r}_{\mathrm{Na}}} = \exp[-i\pi(h + k + \ell)] = (-1)^{\,h+k+\ell}.$$

Thus the NaCl structure factor is

$$F_{hkl} = f_{\mathrm{Cl}}(\mathbf{G}) + f_{\mathrm{Na}}(\mathbf{G})\,(-1)^{h+k+\ell}.$$

Intensity pattern. Equation (2) shows that Na and Cl scatter either *in phase* or *out of phase*, depending on the parity of $h + k + \ell$:

$$F_{hkl} = \begin{cases} f_{\mathrm{Cl}} + f_{\mathrm{Na}}, & \text{if } h + k + \ell \text{ is even,} \\ f_{\mathrm{Cl}} - f_{\mathrm{Na}}, & \text{if } h + k + \ell \text{ is odd.} \end{cases}$$

Since the form factors decrease with increasing $|\mathbf{G}|$ but satisfy $f_{\mathrm{Cl}} > f_{\mathrm{Na}}$ for all scattering angles, the consequences are:

- Reflections with *mixed parity* of (h, k, ℓ) (e.g. 100, 210) are forbidden by the fcc lattice and do not appear.
- Among the allowed reflections (all-even or all-odd), those with $h + k + \ell$ *even* have larger intensity, approximately proportional to $(f_{\text{Cl}} + f_{\text{Na}})^2$.
- Reflections with $h + k + \ell$ *odd* are weaker, approximately proportional to $(f_{\text{Cl}} - f_{\text{Na}})^2$.

Summary of NaCl selection rules. Combining the Bravais-lattice and basis interference rules yields:

Allowed reflections:	h, k, ℓ all even or all odd.
Strong reflections:	$h + k + \ell$ even.
Weak reflections:	$h + k + \ell$ odd.

This intensity alternation is characteristic of the NaCl structure and is commonly used to identify rock-salt–type crystals in powder diffraction.

Finite Crystallite Size and the Scherrer Equation

An ideal infinite crystal produces delta-function–sharp Bragg peaks. Real samples consist of finite-sized crystallites of linear dimension L, and this finite size leads to a broadening of the diffraction peaks. The Scherrer equation provides a simple and widely used relation between the angular peak width and the average crystallite size.

Origin of size broadening. Consider a crystal of finite thickness L along the scattering vector $\mathbf{G}$. The scattering amplitude is proportional to the Fourier transform of the lattice over this finite interval. For a one-dimensional crystal with N layers separated by d, the amplitude is

$$A(q) \propto \sum_{n=0}^{N-1} e^{-iqnd} = e^{-iqd(N-1)/2} \frac{\sin(Nqd/2)}{\sin(qd/2)}.$$

The resulting intensity has a principal maximum at $q = G = 2\pi/d$ and a finite width

$$\Delta q \sim \frac{2\pi}{Nd} \sim \frac{1}{L}.$$

Thus the finite size L leads to an intrinsic broadening of the Bragg peak.

Angular width of a Bragg reflection. In an X-ray diffraction experiment the scattering vector is related to the Bragg angle θ by $|\mathbf{G}| = 2k \sin\theta = \frac{4\pi}{\lambda} \sin\theta$. A small change in scattering vector, Δq, produces an angular width

$$\Delta q = \frac{4\pi}{\lambda} \cos\theta \, \Delta\theta.$$

Combining this with $\Delta q \sim 1/L$ yields

$$\Delta\theta \sim \frac{\lambda}{4\pi L \cos\theta}.$$

Scherrer equation. In practice, the full width at half maximum (FWHM) of a Bragg peak, denoted β, is used. The Scherrer equation is written as

$$L = \frac{K\,\lambda}{\beta\,\cos\theta},$$

where

- L is the average crystallite size perpendicular to the (hkl) planes,
- λ is the X-ray wavelength,
- β is the peak width (FWHM) in radians after instrument correction,
- θ is the Bragg angle,
- K is the dimensionless *shape factor*, typically $K \approx 0.9$.

Experimental X-Ray Diffraction Techniques

The Fourier-space description developed above applies to any diffraction geometry. In practice, the three most widely used techniques are the Laue method, powder diffraction, and single-crystal diffractometry.

Laue method (polychromatic beam). In the Laue technique a stationary single crystal is illuminated with a *white* (polychromatic) X-ray beam containing a continuous spectrum of wavelengths. Different reciprocal-lattice vectors $\mathbf{G}$ satisfy the diffraction condition for different wavelengths, generating a pattern of spots on a detector placed behind or in front of the sample. Key characteristics:

- instantaneous mapping of crystal *orientation*;
- highly useful for aligning single crystals and determining symmetry;
- each diffraction spot corresponds to a specific $\mathbf{G}$ and a specific wavelength in the incident spectrum;
- not suitable for precise lattice-parameter determination.

The Laue method is widely used in crystal growth, device fabrication, and sample alignment prior to single-crystal measurements.

Powder diffraction (Debye–Scherrer geometry). A polycrystalline powder consists of a macroscopic ensemble of randomly oriented crystallites. For a given wavelength λ, each set of planes (hkl) meets the Bragg condition for a subset of crystallites whose orientation satisfies $\mathbf{q} = \mathbf{G}_{hkl}$. As a result, each allowed (hkl) produces a *Debye ring* on the detector. Key features:

- straightforward identification of phases via the set of d_{hkl} spacings;
- intensities correspond to $|F_{hkl}|^2$ but averaged over orientation;
- peak positions give lattice parameters, peak widths can be analyzed via the Scherrer equation;
- insensitive to orientation, requires only powdered material.

Powder diffraction is the standard tool for phase analysis, structural refinement (Rietveld method), and measurement of crystallite size and strain.

Single-crystal diffraction (monochromatic beam). A high-quality single crystal is rotated in a monochromatic X-ray beam while diffraction intensities are recorded for many independent $\mathbf{G}$ vectors. The resulting dataset of I_{hkl} provides the full information needed to determine:

- atomic positions within the unit cell (from F_{hkl}),
- symmetry operations and space group,
- atomic displacement parameters and refined lattice parameters.

Because intensities reflect detailed phase relationships inside the structure factor, single-crystal diffraction enables true three-dimensional structure determination, including refinement of complex molecular structures and subtle distortions.

3 Neutron and Mössbauer Spectroscopy

Neutron Imaging and Spectroscopy

Neutron-based imaging techniques provide a striking demonstration of the unique interaction of neutrons with matter. In contrast to X-rays, which are primarily sensitive to the electron density and thus scale roughly with the atomic number Z, thermal and cold neutrons interact through short-range nuclear forces. As a result, the attenuation of neutron beams exhibits a highly irregular and non-monotonic dependence on Z and even varies dramatically between isotopes of the same element (Fig. 5**a**). A particularly important case is hydrogen, which possesses an unusually large scattering cross section. Materials rich in hydrogen therefore appear strongly absorbing in neutron images, while many metals are comparatively transparent.

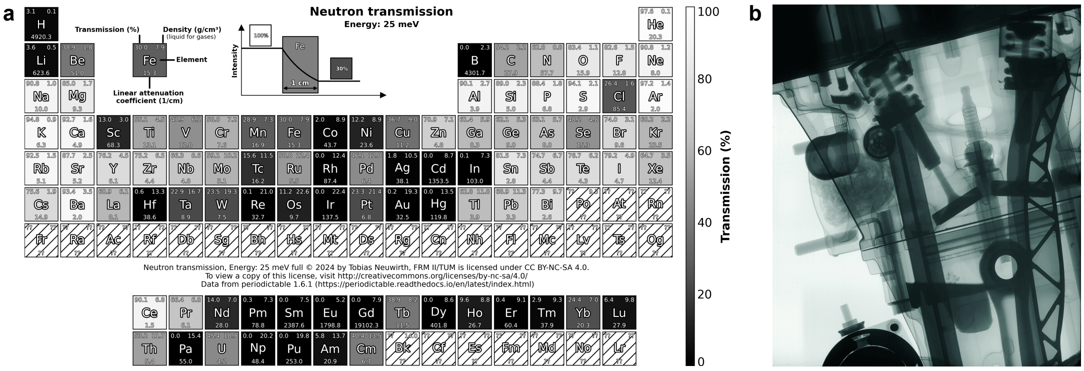

Figure 5: Neutron imaging and scattering. **a** Periodic table highlighting the irregularly varying absorption of thermal neutrons (kinetic energy 25 meV) across the elements. The color indicate strong (black) to light (light grey) neutron absorption. **b** Neutron radiography of a motorcycle engine recorded at the ANTARES beamline of FRM II. Copyright MLZ.

A powerful illustration of this complementary contrast mechanism is neutron tomography. Using a collimated beam of thermal or cold neutrons, a series of radiographs is recorded while the sample is rotated. A tomographic reconstruction then yields a full three-dimensional map of the neutron attenuation coefficient. Neutron tomography has enabled imaging scenarios that are essentially impossible with X-rays. For example, in-operando studies of combustion engines reveal the distribution of water, oil, and fuel inside metal housings that are nearly transparent to neutrons (Fig. 5**b**).

Neutron Optics and Detection

Due to their electrical neutrality neutrons penetrate deeply into matter, experiencing only weak refraction. Classical refractive lenses are therefore of little use. Instead, neutron imaging relies on geometric collimation or pinhole apertures to define the beam, often complemented by neutron guides and supermirror coatings that exploit total external reflection at grazing incidence.

Because neutrons do not ionize matter directly, their detection requires a conversion reaction that produces charged particles. For thermal and cold neutrons this is achieved efficiently using nuclei with large capture cross sections, e.g. ^{3}He, ^{10}B, or ^{6}Li. The reaction products (protons, α particles, or tritons) generate ionization or scintillation light, enabling electronic readout.

For neutron tomography, typical detection schemes employ thin scintillator screens (^{6}LiF/ZnS or ^{6}Li-doped glass) coupled to sensitive optical cameras. Spatial resolutions on the order of tens of micrometers are routinely achieved, and the excellent penetration of neutrons allows imaging of complex, enclosed metallic assemblies.

Elastic Neutron Scattering and Magnetic Structure

Neutron diffraction is formally analogous to X-ray diffraction: a coherent neutron beam with wave vector $\mathbf{k}$ impinges on a sample and is elastically scattered into $\mathbf{k}'$. The condition for constructive interference is the familiar Bragg relation

$$\mathbf{G} = \mathbf{k} - \mathbf{k}', \qquad |\mathbf{G}| = \frac{2\pi}{d_{hkl}},$$

where $\mathbf{G}$ is a reciprocal lattice vector. The measured Bragg peaks reflect the periodicity of the underlying crystal lattice.

A crucial difference to X-rays, however, lies in the interaction mechanism. X-rays couple to the electronic charge density, whereas neutrons interact with individual nuclei through the strong force and additionally carry a magnetic moment. The nuclear interaction provides structural information even for light elements and isotope-sensitive contrast. The magnetic interaction, by contrast, renders neutrons directly sensitive to magnetic lattices. The magnetic lattice may be of various microscopic origins ranging from periodic alignment of individual spins at atomic sites, periodic alignment of magnetic flux quanta (vortices) in superconductors, to periodic alignment of magnetic textures in so-called skyrmions lattices.

Inelastic Neutron Scattering: Excitations and Dispersion

Elastic scattering probes the static arrangement of atoms and spins. In inelastic neutron scattering, by contrast, the neutron exchanges a small amount of energy and momentum with the sample. This process reveals the *dynamics* of the crystal: collective vibrations of the lattice, spin waves in magnetically ordered materials, and other low-energy excitations. When an incoming neutron of wave vector $\mathbf{k}$ and energy E is scattered into $\mathbf{k}'$ and E', the momentum and energy transfers are

$$\mathbf{q} = \mathbf{k} - \mathbf{k}', \qquad \hbar\omega = E - E'.$$

The measured scattering intensity as a function of $(\mathbf{q}, \omega)$ constitutes a map of the elementary excitations in the material.

Collective Excitations At the energy scale of thermal and cold neutrons, matter does not respond through the motion of isolated atoms. Instead, the relevant degrees of freedom are *collective modes*: correlated oscillations that extend over many unit cells.

- **Phonons** are quantized lattice vibrations. The atoms in a crystal vibrate collectively with well-defined frequencies and wavelengths. Neutrons couple to these vibrations through the nuclear interaction.

- **Magnons** are quantized spin waves in magnetically ordered systems. Here the neutron interacts through its magnetic moment. Disturbing the ordered spin structure by flipping or twisting a spin creates a propagating excitation whose frequency (or energy) depends on the wavelength of the disturbance.

Dispersion Relations A central concept in the interpretation of inelastic scattering is the *dispersion relation* $\omega(\mathbf{q})$ of an excitation. It specifies how the frequency (or energy) of a collective mode depends on its wavelength. Typical qualitative behaviours include:

- acoustic phonons: frequency increases approximately linearly with $|\mathbf{q}|$ near the Brillouin-zone centre;
- optical phonons: finite frequency at $\mathbf{q} = 0$;
- ferromagnetic magnons: quadratic dispersion at small $|\mathbf{q}|$;
- antiferromagnetic magnons: approximately linear dispersion near the ordering wave vector.

Experimental realization

In practice, neither the neutron momentum nor the momentum transfer is measured directly. Instead, $\mathbf{q}$ is reconstructed from the well-defined geometry and/or timing information of the instrument.

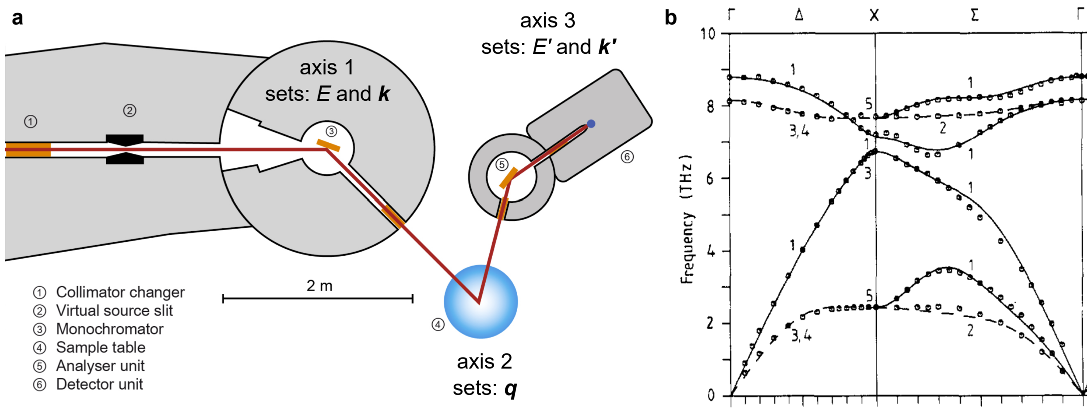

Figure 6: Inelastic Neutron scattering. **a** Schematic of a so-called triple axis spectrometer. Copyright MLZ. **b** Phonon dispersion of GaAs measured by inelastic neutron scattering. Reproduced with permission from Strauch and Dorner, J. Phys. Condens. Matter 2 1457 (1990). Copyright 1990 IOP Publishing.

Triple–axis spectroscopy. A triple–axis spectrometer employs three independently rotatable axes (Fig. 6**a**). (i) a monochromator that selects the incident momentum $\mathbf{k}$, (ii) the sample, which sets the scattering angle, and (iii) an analyser that selects $\mathbf{k}'$. By rotating these components, one chooses the magnitudes

$$|\mathbf{k}| = \frac{2\pi}{\lambda_i}, \qquad |\mathbf{k}'| = \frac{2\pi}{\lambda_f},$$

and their relative angle. The selected $\mathbf{q}$ follows directly from the calibrated mechanical angles. Scans through $\mathbf{q}$ are performed by rotating the sample and detector arm, while varying $\mathbf{k}$ or $\mathbf{k}'$ tunes the energy transfer $\hbar\omega = E - E'$. The monochromator and analyser are typically single crystals (such as crystal of pyrolitic graphite) that diffract well defined energy and momenta of the beam according to Bragg's law. Figure 6**b** shows exemplarily the phonon dispersion $\omega(\mathbf{q})$ of GaAs as measured by inelastic neutron scattering.

Time–of–flight spectroscopy. In a time–of–flight (TOF) experiment the incident beam contains a broad wavelength distribution. The initial momentum $\mathbf{k}$ is determined for each neutron from its arrival time at the sample position. After scattering, the neutron is detected at a known flight path and at a specific detector angle, yielding both $|\mathbf{k}'|$ (from the flight time) and its direction (from the pixel position). Each detection event thus produces an unambiguous point in $(\mathbf{Q}, \omega)$ space. Typical velocities of thermal neutrons are 2200 m/s, while typical length scales of the detector geometry are several meters resulting in timescales of µs, well within reach of standard electronic detectors.

Mössbauer Spectroscopy

Historical Background

In 1957, Rudolf Mössbauer, then a doctoral student at the Technische Universität München, discovered that γ-rays emitted by nuclei embedded in a solid can be absorbed by identical nuclei without energy loss to recoil. This was unexpected, because a free nucleus must recoil after emission, shifting the γ-ray out of resonance. Mössbauer showed that in a crystal the recoil momentum can be taken up by the entire lattice, enabling resonant, recoil-free absorption of γ-radiation with extraordinarily narrow linewidth. The discovery was soon confirmed by high-precision measurements, including the famous Pound–Rebka experiment (1960) testing gravitational redshift. Only four years after this doctoral work, Mössbauer received the 1961 Nobel Prize in Physics, one of the fastest recognitions in the history of the award.

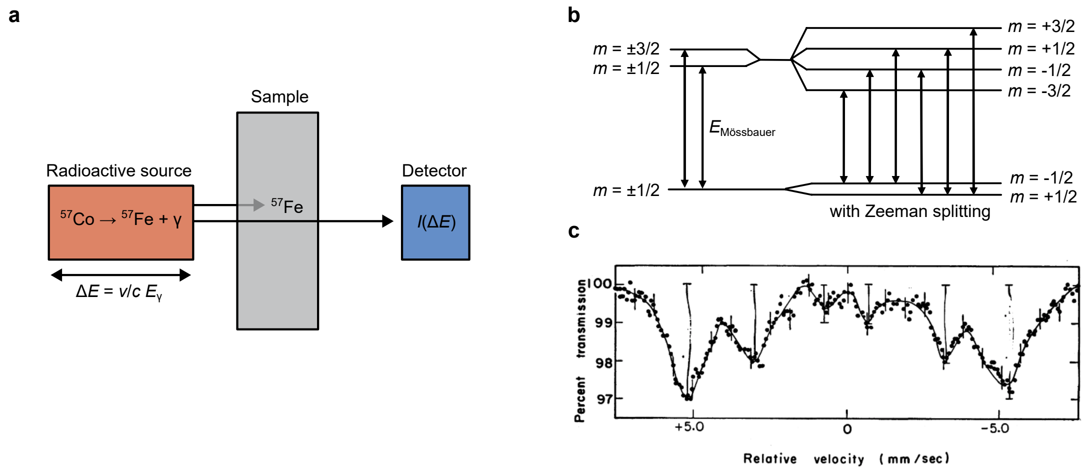

Figure 7: Mössbauer spectroscopy. **a** Schematic of Mössbauer spectroscopy. A radioactive source emits gamma radiation produced via the excited state of a specific isotope. The radiation is resonantly absorbed by the *same isotope* in the sample. The energy is precisely modulated by moving the source (Doppler effect). The transmission resonances are detected by a photon counting detector. In this example, cobalt decays via electron capture into an excited state of iron which decays into the ground state via emission of a gamma quant. **b** Level diagram of ^{57}Fe with Zeeman splitting from magnetization. Transitions (arrows) follow the usual dipole selection rule $\Delta m = \pm 1, 0$. **c** Mössbauer absorption spectrum of ^{57}Fe showing the typical six absorption lines. Reproduced with permisison from Science 141, 1035-1038 (1963). Copyright 1963 AAAS.

Basics of the Mössbauer Effect

A free nucleus emitting a γ-ray of energy E_γ must recoil with energy

$$E_R = \frac{E_\gamma^2}{2Mc^2},$$

which for the 14.4 keV transition of ^{57}Fe is of order 10^{-3} eV. In a solid, however, the lattice can take up momentum only by creating phonons, whose energies are quantized in units $\hbar\omega_{\text{phonon}}$ of order meV. Since

$$E_R \ll \hbar\omega_{\text{phonon}},$$

the recoil energy is too small to create even a single (optical) phonon. Phonon creation is thus forbidden, and the momentum is absorbed elastically by the whole lattice. Such recoil-free events occur with probability $f = e^{-2W}$, the Lamb–Mössbauer factor.

Because these photons retain their natural, extremely narrow linewidth, the resulting absorption spectra resolve tiny hyperfine interactions. Three effects are relevant: (i) the isomer shift, which probes s-electron density at the nucleus; (ii) quadrupole splitting due to electric field gradients for nuclei with $I \geq 1$; and (iii) magnetic hyperfine splitting arising from internal magnetic fields.

Experimental Implementation: Doppler Modulation

To scan the γ-ray energy across the nuclear resonance, the Mössbauer source is moved with a precisely controlled velocity v (typically a few mm/s). The resulting Doppler shift

$$\Delta E = \frac{v}{c}\, E_\gamma$$

is sufficient to tune the photon energy by an amount comparable to the hyperfine splittings. A schematic experimental setup is shown in Fig. 7**a**: a driven source illuminates an absorber containing the isotope of interest, and a detector records the transmitted intensity as a function of Doppler velocity. The measured transmission spectrum reveals the hyperfine splitting at the absorber nucleus.

Example: Sextet Spectrum of α-Fe

The classic example is elemental α-iron, where the ^{57}Fe nucleus undergoes a 14.4 keV transition. The nuclear levels have spins

$$I_g = \tfrac{1}{2}, \qquad I_e = \tfrac{3}{2}.$$

In ferromagnetic α-Fe, an internal hyperfine field of approximately 33 T produces Zeeman splitting into 2 sublevels of the ground state and 4 sublevels of the excited state (Fig. 7**b**). Allowed transitions obey $\Delta m = 0, \pm 1$, resulting in six absorption lines: the characteristic Mössbauer sextet shown in Fig. 7**c**.

The spacing of these six lines yields the hyperfine magnetic field, while their positions and intensities encode information on chemical environment, symmetry, and magnetic domain orientation at the iron sites. This makes the α-Fe spectrum a textbook example of how Mössbauer spectroscopy provides site-specific insight into hyperfine interactions with exceptional energy resolution.